MIL-T-58082A(AV)
5 April 1974
Superseding
MIL-T-58082(MO)
29 January 1965

MILITARY SPECIFICATION

TESTER, PYROMETER AND THERMOCOUPLE, FIELD, TYPE N-3A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pyrometer and thermocouple field tester, designated N-3A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet
TT-L-32	Lacquer, Cellulose Nitrate, Gloss, for Aircraft Use
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

MILITARY

MIL-B-18	Batteries, Dry
MIL-W-76	Wire and Cable, Hookup, Electrical Insulated
MIL-P-116	Preservation, Packaging, Methods of
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-101	Preservation, Packaging, and Packing Materials, Test Procedures

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-808	Finishes; Protective and Codes for Finishing Schemes for Ground and Support Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-889	Dissimilar Metals
MS3106	Connector, Plug, Electric, Straight
MS20257	Hinge, Continuous
MS25165	Knob, Pointer, Illuminated, Small

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design.--The tester shall be designed for testing aircraft temperature indicators of both the ratiometer (resistance thermometer) and the thermocouple types. It shall incorporate a direct current resistance measuring section suitable for measuring the resistance of the thermocouples and associated leads installed in aircraft. Design shall be reasonably simple, to permit overhaul or repair without requiring special tools and fixtures. Layout and assembly of the tester and its

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the currentversion before use.

components shall be generally as shown on figures 1 and 2, to facilitate ease of maintenance in accordance with MIL-S-8512. All rheostat and selector-switch control knobs shall be readily accessible and so designed that they can be operated by personnel wearing gloves. Weight of the tested completely assembled and including accessories, but without batteries, shall not exceed 20 pounds. Requirements specified herein are detailed only to the extent necessary to obtain the desired mechanical and electrical performance in accuracy of indications.

3.1.1 First Article.--Unless otherwise specified (see 6.2), as soon as practicable after award of contract or order, and prior to starting regular production, one tester, and its packaging, as applicable, shall be submitted for examination and testing to determine conformance with this specification (see 4.2.1). A first article sample is defined as a production line item which the manufacturer willsubsequently produce in fulfillment of the contract or order. Approval of the first article sample by the procuring activity shall not relieve the contractor of his responsibility to supply testers conforming to this specification. Any change or deviation from the approved first article sample shall be subject to prior approval of the contracting officer.

3.2 Materials.--Materials shall be as specified herein. When specifications and standards are not specifically designated, selection of materials and processes shall be in accordance with MIL-STD-143. All materials and component parts shall be new and unused.

3.2.1 Substitution of materials.--Materials not conforming to the specifications referenced herein shall not be substituted unless such substitutions are specifically approved by the contracting officer. Substitutions will be approved only if it is clearly demonstrated that an improvement in operating characteristics, a saving in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice of reliability can be accomplished thereby, and that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Requests for approval of such substitutions shall be accompanied by completely detailed supporting data which clearly illustrate the use of the proposed substitute materials.

3.3 Construction.--

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

3.3.1 Standard parts.--AN or MS standard parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawings by their part numbers. In applications for which no suitable corresponding AN or MS part is in effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

3.3.2 Screw threads.--Unless otherwise specified, the threads of all machine screws 0.060-inch or larger in diameter shall conform to MIL-S-7742.

3.3.3 Interchangeability.--Components shall be interchangeable (without requiring modification for replacement) with similar components furnished under the same contract or order.

3.3.4 Soldering.--Soldering shall be in accordance with MIL-S-6872. There shall be no damage to adjacent parts. Sufficient heat shall be applied for solder to flow freely and completely over the joints; there shall be no evidence of "cold joints".

3.3.5 Protective finishes.--Materials subject to deterioration when exposed to climatic and environmental conditions during service usage, shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements specified herein. Protective coating that will crack, chip, or scale with age or extremes of climatic or environmental conditions shall not be used.

3.3.5.1 Metals.--Metals shall be of the corrosion-resistant type or suitably protected to withstand the environmental conditions specified in 4.4.9. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

3.3.5.2 Aluminum alloy.--Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except under these conditions:

(a) Aluminum surfaces which require bonding shall be provided with a means to insure a low-resistance electrical path.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

(b) Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

(c) If the primary purpose of the anodizing is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used.

(d) If abrasion resistance is a factor, chemical films shall not be used.

3.3.5.3 Ferrous alloys.--Ferrous alloy parts shall be protected to insure their function to meet the performance requirements of the tester.

3.4 Components.--

3.4.1 Case.--The case (see fig. 1) shall be 6 inches deep, 10-1/2 inches wide, and 18 inches long. It shall be of 0.081-inch-thick, half-hard aluminum conforming to QQ-A-250/8. The case shall have four feet on the hinged side and four feet on the side opposite the lid. The feet shall be formed by dimpling or other suitable means.

3.4.1.1 Lid.--A detachable lid shall be secured to the case by a hinge conforming to MS20257. Two sturdy corrosion-resistant steel supports shall be provided to relieve hinge strain when the lid is open.

3.4.1.2 Fasteners.--Metal truck-type, pull-down latches shall secure the lid when closed.

3.4.1.3 Case finish.--Both inside and outside surfaces shall be anodized in accordance with MIL-A-8625. The outside surface shall be in accordance with F-300 DG of MIL-STD-808, except that the two coats of final film shall conform to TT-L-32. The anodized finish shall match color No. 13538 of FED-STD-595.

3.4.1.4 Handle.--A hinged metal handle shall be located in the optimum position with respect to weight distribution. Handle shall provide clearance from the case sufficient to pass a block 1-7/8 by 4-3/8 inches in cross section and with rounded edges. Grip portion of handle shall be of nonmetallic material and shaped to fit the hand comfortably.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

3.4.1.5 Battery compartment.--A compartment shall be provided in one end of the case to house a BA-35 battery conforming to MIL-B-18/13 and a BA-36 battery conforming to MIL-B-18/14. Batteries are not to be supplied. The compartment shall be so constructed that the batteries can be rigidly mounted, and when a battery deteriorates there is room for its removal without damage to the tester. The compartment shall be marked to indicate the type of battery and the wiring connections required. Top side of the compartment shall have a hinged door or removable panel for access to the batteries. This panel or door shall be flush with the instrument panel in the case.

3.4.2 Panel.--The tester shall incorporate a metal or phenolic panel for mounting the selector switches, rheostats, voltmeter, and resistance units. All component parts of the tester, except accessories and batteries, shall be mounted on the panel to form an assembled unit. Panel arrangement shall be substantially in accordance with figure 2.

3.4.2.1 Markings.--The designation of switches and indicator shall be etched or engraved in the panel to a depth of 0.010-inch, plus or minus 0.005-inch. Lettering shall be 0.093-inch high by 0.015-inch line width. Numerals designating the switches and indicator shall be 0.125-inch high by 0.020-inch line width. The markings shall be filled with white enamel or other suitable material. Text and approximate location of markings shall be as shown on figure 2.

3.4.3 Meter.--The standardizing meter shall conform substantially to figure 3. Unless otherwise specified herein, the meter shall conform to the basic requirements of MIL-M-10304 for sealed meters. The meter mechanism shall be sufficiently strong and rigid to enable the meter to withstand, without damage, ordinary transportation, handling, and service.

3.4.3.1 Jewels and pivots.--Moving elements shall turn on hardened, accurately ground and highly polished pivot bearings, highly polished first quality sapphire bearings, or other materials that have been approved as alternates to precious gems.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the currentversion before use.

3.4.3.2 Balancing.--The moving element shall be balanced by approved means (such as threaded lock, split nuts, or a wire helix moving along a cross arm and counterweight arm), with easily accessible provisions for balancing. (Note: The use of solder, shellac, or similar means for holding threaded balance nuts prevents ready balancing and will not be approved.)

3.4.3.3 Damping factor.--The moving element of the meter shall have an efficient damping arrangement to prevent excessive pointer oscillation. The damping factor of the meter shall be not less than 2.5 (see 6.3.2).

3.4.3.4 Scale length.--Scale length of the meter shall be not less than 3-1/2 inches as measured on the outer arc of the graduations.

3.4.3.5 Pointer.--A knife-edge pointer shall permit accurate readings at the usual distance from which viewed. The pointer shall be of the channeled or equally rigid construction and suitable for the application in which it is used. The knife edge of the pointer shall have a durable dull black finish.

3.4.3.6 Cover glass.--The cover glass shall be flat and free from discoloration, scratches, and striae which interfere with normal reading of the meter.

3.4.3.7 Zero adjuster.--A zero adjuster shall provide for a range of adjustment, above and below the zero point, of not less than 3 percent of the scale length. The adjuster shall rotate through 360 degrees without damage to any part of the meter, and without becoming inoperative. The adjuster shall have sufficient friction to prevent it from shaking loose and changing adjustment when subjected to the tests specified in 4.4.5. The adjuster shall be accessible from the front of the meter.

3.4.3.8 Response time.--After momentary full scale deflection, the meter pointer shall come to rest within plus or minus 0.2-percent of scale length within 2 seconds, plus or minus 1 second.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

3.4.3.9 Dial.-- The meter dial shall be made of stiff material that will not warp or deteriorate under extreme climatic conditions. It shall be firmly secured to the mounting, and readily removable from the meter. The dial shall incorporate a mirror near the edge of the scale arc and directly below the knife edge of the pointer, to reduce errors in reading due to parallax. Dial markings shall conform to figure 3. Reference marks and minimum and maximum voltage values shall be provided. Dial finish shall be as specified in table I.

TABLE I. -- Dial markings on meter.

Dial markings	Color	Height or length, ± 0.005-inch	Width of line, ± 0.002-inch
Dial background	White		
Scale arc	Black		0.020
Graduation at "0"	Black	0.250	0.008
Graduation at 11.25 and 22.50	Black	0.125	0.008
Graduation at 14.25 and 28.50	Red	0.250	0.008
All letters	Black	0.093	0.020
All numerals	Black	0.093	0.020

3.4.4 Temperature selector switch No. 2.--A temperature-selector switch (see fig. 2) shall select the various temperature calibration points shown on figure 4. Switch shall incorporate a positive indexing mechanism. It shall be of the shorting type and shall be constructed with silver contacts, collector rings, and brush shoes. Brushes shall have sufficient pressure to maintain proper contact between brush shoes, collector rings, and contact points to hold contact resistance within 0.002-ohm. The switch shall be of sturdy construction and designed to afford maximum reliability of results and long life under service conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

3.4.4.1 Dial.--The dial (see fig. 4) shall be of metal and securely attached to the panel, or fabricated directly on the panel. Dial markings shall be durable to withstand usage encountered in service. All letters, numerals, graduations, and circles, shall be engraved or etched to a depth of 0.010-inch, plus or minus 0.005-inch, and filled with white enamel or other suitable material as listed in table II. Dial finish shall be in accordance with F-300 of MIL-STD-808. Colors shall be in accordance with FED-STD-595, except that the two coats of final film shall conform to TT-L-32.

TABLE II. -- Dial markings of temperature selector switch.

Dial markings	Background color No.	Height or length, ± 1/64-inch	Width of line, ± 0.005-inch
Iron-constantan (sector)	Yellow 33538		
Copper-constantan (sector)	Red 31136		
Chromel-alumel (sector)	Green 34108		
Inner portion of dial	Black 37038		
"OFF" (sector)	Black 37038	1/8	0.020
Letters	White	3/32	0.015
Graduations	White	1/4	0.030
Numerals	White	1/8	0.020
Circle	White	3-1/4 dia.	0.20

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the currentversion before use.

3.4.5 Selector switch No. 8.--A two position shorting-type selector switch shall be provided as shown on figure 2. The switch, when set in the "Left and Single" position, shall test resistance thermometer indicators of the single-engine type, and the left-hand scale of dual-type indicators. The switch, when set in the "Right" position, shall test the right-hand scale of dual-type indicators. When testing dual-type indicators, the selector switch shall provide for a bulb-circuit load of approximately 100 ohms across the right-indicator movement when testing the left-indicator movement, and vice-versa. The switch shall be constructed with silver contacts, collector rings, and brush shoes. Brushes shall have sufficient pressure to maintain proper contact between brush shoes, collector rings, and contact points, to hold contact resistance within 0.002-ohm.

3.4.6 Resistance voltage switch No. 7.--An eight-position resistance and voltage selector switch shall be provided as shown on figure 2. The switch shall be suitable for connecting 2 ohms, plus or minus 0.05-ohm, 8 ohms, plus or minus 0.05-ohm, or 22 ohms, plus or minus 0.05-ohm, in series with the thermocouple indicator under test. The 2-ohm, 8-ohm, and 22-ohm resistance units shall be incorporated in the tester in series with the thermocouple-indicator test-load circuits. The switch without any series resistance shall also provide means for testing the thermocouple indicator and its external circuit on aircraft and for selecting either 12- or 24-volt, as required for testing resistance-type thermometers and for selecting the thermocouple lead resistance check circuit.

3.4.7 Safety switch.--A battery safety switch with features for sealing against the entrance of moisture, dust, and other foreign material shall be incorporated in the tester. The switch shall be operated by the lid of the tester case in a manner that when the lid is closed, the battery circuits are "OFF" and when the lid is open, the battery circuits are "ON".

3.4.8 Rheostat controls (Nos. 5 and 6).--Rheostat controls shall be provided on the panel as shown on figure 2. No. 5 rheostat shall be for coarse adjustment, and No. 6 rheostat for fine adjustment. With the fine adjustment positioned at the extreme counterclockwise position, it shall not be necessary to rotate the coarse adjustment in the clockwise direction more than 95 percent of its resistance range to obtain 50 millivolts output at the standardizing jacks with loads of either 17 ohms or 42 ohms. The rheostats shall provide power regulations as indicated on the standardizing meter for resistance-thermometer and thermocouple-thermometer test circuits.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

After the standardizing meter pointer has been set on the red line, no further adjustment shall be required when selector switch No. 2 is rotated over the range of temperature test points provided for the respective thermocouple thermometer test circuits. The rheostat adjustments shall be sufficiently fine to permit setting of the standardizing meter pointer on the red line of the thermocouple-thermometer test circuit, and within 0.020-inch of the setting point on the resistance-thermometer test circuit.

3.4.9 Thermocouple resistance lead circuit and controls.--A resistance bridge circuit shall be incorporated in the tester for facilitating the measurement of 2-ohm, 8-ohm, and 22-ohm thermocouple lead circuits in aircraft. Switch No. 7 when placed in the "Lead Resistance Check" position shall electrically energize the resistance bridge from the 1.5-volt battery. Switch No. 3 shall control the switching of precision resistance in the bridge as required for measuring the 2-ohm, 8-Ohm, and 22-ohm thermocouple lead circuit. Resistance control No. 4 shall balance the bridge and register the error existing in the particular lead circuit being measured over a range of minus 1 ohm to plus 1 ohm. The calibration markings shall be substantially as shown on figure 2 and the minor and major graduations shall register in increments of 0.05-and 0.10-ohm respectively. The standardizing meter shall be employed as a null galvanometer for registering bridge balance. The bridge shall be balanced when the meter pointer is directly over the "Zero" graduation. The bridge circuit shall measure the resistance of applicable thermocouple lead circuits within a tolerance of plus or minus .02-ohm. Switch No. 3 may be in the form of three individual pushbutton switches.

3.4.10 Control knobs.--Each switch and resistance control, with the exception of the temperature selector switch No. 2, shall be furnished with a control knob conforming to MS25165. The temperature-selector switch shall have a bar-type knob and pointer. The knob shall be approximately 2-1/2 inches long and of a shape that will clearly indicate the test points on the dial.

3.4.11 Electromotive force test circuit.--Two pin jacks shall be mounted on the panel as shown on figure 2. The jacks shall be identified with plus and minus marks on the panel. A cover plate, easily removable for access to the jacks, shall cover the jacks when not in use, and shall be marked as shown on figure 2. The jacks shall be connected into the tester circuits in a manner that the electromotive force values shown in 4.4.2 will be indicated on the master equipment specified in 4.3.3 when the indicator load, as specified in table V, is connected across the thermocouple lead clips, and switch No. 7 is in the 8-ohm position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

The plus and minus relationship of the jacks shall apply to positive electromotive force values as specified in table VI.

3.5 Sealing.--The instrument case, meter, panel, and control shafts shall be adequately sealed to meet the requirements specified herein. Gasket materials, where used, shall be fungus inert, and shall be secured in a manner not to be damaged by removal of the mating parts. In general, the following types of material shall be used:

(a) Cellulose acetate.
(b) Nylon
(c) Rubber (natural or synthetic).

3.6 Accessories.--

3.6.1 Thermometer.--The liquid-in-glass thermometer shall be Taylor Instrument Companies' model P/N X-108, or equal. Clips shall be provided in the lid of the tester case for storage of the thermometer when it is not in use.

3.6.2 Thermocouple test leads. Two test leads, each approximately 5 feet long, shall be provided for making thermocouple-indicator tests as shown on figure 1. The leads shall be No. 12, B and S flexible copper wire, stranding 65/30, approximately 3/64-inch rubber insulation, color optional. One end of each lead shall be securely attached to the applicable electrical circuit within the tester. The other end of each lead shall be provided with universal test clips and rubber insulators. The leads shall be identified with plus and minus markings by means of tags permanently attached to the test leads.

3.6.3 Resistance thermometer test lead.--A resistance thermometer test lead (see fig. 1) for making resistance-thermometer indicator tests, shall be approximately 5 feet long and shall comprise five conductors, each of which employs stranded copper wire, size No. 20 AWG, conforming to type MW of MIL-W-76. The conductors shall be grouped and protected by insulation sleeving conforming to MIL-I-7444. One end of the lead shall be secured to the applicable electrical circuits within the tester. The other end of the lead shall be provided with an electrical-connector plug conforming to MS3106A-14S-5S. The sequence of wiring between the tester and the connector plug conforming to MS3106A-14S-5S shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

Connector pins	Electrical functions
A	Plus 28.5 volts.
B	Decade resistance (left and single).
C	Minus decade resistance and ground.
D	Plus 14.25 volts.
E	Decade resistance (right).

3.6.4 Adapter test leads.--Detachable leads for testing various types of resistance-thermometer indicators shall conform to figure 5. Leads shall be stored in the accessory compartment of the tester shown on figure 1. Unless otherwise specified, wire used in the construction of the test leads shall conform to type MW, size No. 20 AWG of MIL-W-76. Electrical connectors shall be in accordance with MIL-C-5015. Lead wires between the electrical connectors shall be protected with plastic tubing conforming to MIL-I-7444. Ends of the tubing shall be wrapped with tape and secured with electrical connector clamps. The electrical connectors shall be furnished without coupling nuts to permit the connecting and disconnecting of the adapter leads for instrument testing more rapidly.

3.6.5 Wiring.--Unless otherwise specified, wiring used in the tester shall be in accordance with MIL-W-76. All wiring, insofar as practicable, shall be distinctly color-coded to facilitate servicing of the tester.

3.6.5.1 Wiring diagram.--A wiring diagram, permanent in nature, and as large as space permits, shall be secured to the inner surface of the tester case under the instrument panel. The diagram shall contain information pertinent to the electrical circuits and component parts of the tester, for reference during servicing operations.

3.7 Performance.--The tester shall be capable of meeting the test requirements specified in section 4 when subjected to the following conditions:

(a) Temperatures ranging from minus 54°C. to plus 71°C. (see 4.4.9.1 and 4.4.9.2).
(b) Fungus growth as encountered in tropical climates (see 4.4.9.3).
(c) Impact shocks of 25 gravity (g) (see 4.4.9.4).

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

3.8 Identification of product.--Tester assembly and its components shall be identified in accordance with MIL-STD-130, and shall include the manufacturer's part number.

3.9 Workmanship.--The tester, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding, brazing, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges. The tester shall be thoroughly cleaned after final inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests.--Examinations and tests shall be classified as (1) first article and (2) conformance.

4.2.1 First article.--The first article tester shall be evaluated for ease of maintenance, servicing, operation, and provisions to prevent accumulation of dirt and other matter which could hinder operation. It shall also be subjected to the examinations specified in 4.2.2.1 and to the tests specified in 4.4. A preproduction package shall be fabricated per section 5 and tested per 4.5.1 when the level of packaging specified is A or B.

4.2.2 Conformance.--Each tester shall be subjected to the examinations specified in 4.2.2.1 and to the tests specified in 4.2.2.2.

4.2.2.1 Examination.--The tester shall be examined to determine:

(a) Compliance with this specification with respect to materials, workmanship, and markings.

(b) Correctness of assembly.

(c) Adequacy of wiring connections.

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:00Z
Check the source to verify that this is the currentversion before use.

4.2.2.2 Tests.--Each tester shall be subjected to the tests specified in 4.4.1 through 4.4.8.

4.3 Test conditions.--

4.3.1 Atmospheric.--Unless otherwise specified, tests shall be made at atmospheric pressure of approximately 29.92 inches of mercury (hg) and at a temperature of 25° Centigrade (C), plus or minus 2°C. If tests are made with atmospheric pressure or temperature different from these values, allowances shall be made for the difference from the specified conditions.

4.3.2 Tapping.--Unless otherwise specified, the tester shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Master equipment.--The equipment used to determine the accuracy of the millivolt (mv) output and resistance values of the tester shall have a certified accuracy within plus or minus 0.01-percent on resistance measurements, and plus or minus 5 microvolts between 0 and 16 mv, and plus or minus 7 microbolts between 16 and 50 mv on the mv measurements at each calibration point.

4.4 Tests.--

4.4.1 Decade resistance.--A precision Wheatstone bridge shall be connected between pins "B" and "C" on the resistance thermometer test lead as shown on figure 1. Switch No. 8 shall be placed in the "Left and Single" position, and switch No. 2 in the position indicated in tables III and IV. The resistance values between pins "B" and "C" shall correspond to the values specified in tables III and IV within plus or minus 0.05-ohm at any test point. With the Wheatstone bridge connected to pins "E" and "C" on the resistance-thermometer plug, and switch No. 8 in the "Right" position, the resistance values between pins "E" and "C" shall correspond to the values specified in tables III and IV within plus or minus 0.05-ohm at any test point. The resistance values specified in tables III and IV are based on the absolute ohm.

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:00Z
Check the source to verify that this is the currentversion before use.

TABLE III.--Decade resistance (90.38 ohms).

Position of switch No. 2	Resistance (ohms)	Position of switch No.2	Resistance (ohms)
-70	68.27	+ 80	120.36
-50	74.24	+100	128.85
-30	80.56	+120	137.78
-10	87.04	+150	151.91
0	90.38	+200	177.95
+10	93.80	+250	208.00
+30	100.91	+300	242.70
+50	108.39	---	----

TABLE IV.--Decade resistance (50.00 ohms).

Position of switch No. 2	Resistance (ohms)	Position of switch No. 2	Resistance (ohms)
-50	37.95	---	-----
0	50.00	+200	116.50
+50	63.55	+250	139.00
+100	79.00	+300	164.50
+150	96.50	----	-----

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

4.4.2 Electromotive force values.--Switch No. 7 shall be placed in the 8-ohm position, and a load as indicated in table V, connected across the thermocouple lead clips. The meter pointer shall be adjusted to, and maintained on, the red line during test. The electromotive force (emf) values of the tester shall be measured across the tester standardizing pin jacks by means of the master equipment specified in 4.3.3. The values obtained for the various positions of switch No. 2 shall correspond to the values specified in table VI within plus or minus 0.04-mv, between 0 and 20 mv, and plus or minus 0.06-mv between 20 and 42 mv. The millivoltage at the pin jacks shall not change more than 0.042-mv when switch No. 7 is changed from the 8-ohm position to the 22-ohm position, and switch No. 2 is in the 1,000°C. chromel-alumel position. The millivoltage at the pin jacks shall not change more than plus or minus 0.05-mv when the 17-ohm and the 42-ohm leads are varied respectively over ranges of 15 to 19 ohms and 35 to 49 ohms.

TABLE V.--Thermocouple load values.

I-C	Iron-constantan	20°C. ref junction. 17-ohm load
C-C	Copper-constantan	20°C. ref junction 17-ohm load
C-A	Chromel-alumel	0°C. ref junction. 42-ohm load

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

TABLE VI.--Electromotive force values.

Switch No. 2 test point	°C.	Millivolts	Switch No. 2 test point	°C.	Millivolts
I-C	-50	-3.57	C-C	100	3.49
I-C	0	-1.05	C-C	150	5.91
I-C	+20	0.00	C-C	200	8.50
I-C	50	+1.61	C-C	250	11.22
I-C	100	4.35	C-C	300	14.07
I-C	150	7.14	C-C	350	17.03
I-C	200	9.94	C-A	0	0.00
I-C	250	12.74	C-A	200	8.13
I-C	300	15.51	C-A	400	16.39
I-C	350	18.27	C-A	500	20.64
C-C	-50	-2.59	C-A	600	24.90
C-C	0	-0.79	C-A	700	29.14
C-C	+20	0.00	C-A	800	33.31
C-C	50	+1.25	C-A	1,000	41.31
---	---	-----	C-A	1,200	48.89

4.4.3 Clip lead resistance.--Switch No. 2 shall be set to the zero reference line on the chromel-alumel sector. Switch No. 7 shall be set in each of the following positions: 0, 2, 8, and 22 ohms, and the resistance measured across the thermocouple-thermometer lead clips with the master equipment specified in 4.3.3. The measured resistance shall be within plus or minus 0.04-ohm of 0, 2, 8, and 22 ohms, respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

4.4.4 Position error.--The rest of the moving element shall be checked by applying sufficient power to the meter to bring the pointer exactly in the center of the "red" adjust line. When the face of the meter is tilted 30 degrees in any direction from a horizontal plane, the pointer shall remain on the "red" adjust line, and no portion of the white background of the dial shall be visible between the pointer and the adjust line.

4.4.5 Zero adjustment.--The zero adjustment shall be carefully examined to determine compliance with 3.4.3.7.

4.4.6 Voltage test.--Sufficient voltage shall be applied to the meter to bring the pointer to the voltage marking on the dial. The laboratory voltmeter used to determine the voltage shall have an accuracy within plus or minus 0.2-percent. The meter shall be adjusted to the 28.50-14.25-volt mark, and the 22.50-11.25-volt mark, respectively. The voltage shall be measured between pins "A" and "C" of the resistance-thermometer test lead for the 28.50-22.50 voltage values, and between pins "D" and "C" for the 14.25-11.25 voltage values. These voltage values shall be measured with a load resistor of 500 ohms, plus or minus 50 ohms, placed across pins "A" and "C" or "D" and "C", as applicable. The voltage markings specified on the meter dial as shown on figure 3 shall correspond to the values obtained on the laboratory meter within plus or minus 0.5-volt.

4.4.7 Adapter leads.--Adapter leads shall be checked for continuity of wiring, and shall be tested by the application of 500 volts of commercial frequency between any two nonconnected terminals, and between terminals and other exposed metal parts. Any indication of flow of current as evidenced by the lighting of an electric bulb or a reading on a suitable electrical meter in the circuit, shall be considered an indication of failure.

4.4.8 Thermocouple lead resistance measuring circuit.--Switch No. 7 shall be set in the "lead resistance check" position. Switch No. 3 shall be set in each of the following positions: 2, 8, and 22 ohms, and with resistance standards of 2, 8, and 22 ohms respectively connected to the thermocouple-thermometer lead clips. Adjustment No. 4 shall be rotated until the bridge is "nulled", as registered by the standardizing meter, for each resistance check. The error registered by adjustment No. 4 for the 2-ohm, 8-ohm, and 22-ohm resistance check positions shall not differ by more than plus or minus 0.02-ohm from the true resistance values of the respective standards being measured. The test shall be repeated with resistance standards of 1.8 ohms, 2.2 ohms, 7.5 ohms, 8.5 ohms, and 22.8 ohms. The measured tolerance listed above shall apply.

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:00Z
Check the source to verify that this is the currentversion before use.

4.4.9 Environmental.--The tester shall be subjected to the environmental tests conducted in accordance with the following:

4.4.9.1 Low temperature.--The tester shall be placed within a test chamber capable of maintaining an internal temperature of minus 54°C., plus or minus 2°C., for a period of 24 hours. At the end of the 24-hour period, the temperature shall be raised to minus 40°C., plus or minus 2°C., for a period of 4 hours. At this temperature, the tester shall be subjected to the tests specified in 4.4.1, 4.4.2, and 4.4.8. The decade-resistance values obtained shall not vary more than 0.4-percent from the initial resistance values specified in tables III and IV. The emf values obtained shall not vary more than 0.08-mv between 0 and 20 mv, and not more than 0.12-mv between 20 and 50 mv from the initial values specified in table VI. Resistance standards of 2, 8, and 22 ohms, maintained at room temperature, shall be measured by means of the thermocouple lead resistance measuring circuit. The error registered by adjustment No. 4 shall not differ by more than plus or minus 0.05-ohm from the true resistance values of the standards. While the tester is at minus 40°C., the switches and rheostat shall be operated to determine that no sticking occurs and that they operate freely. An external power source may be used for this test. After the tester has returned to room temperature, measurements shall be conducted to determine the calibration accuracy. There shall be no damage which would adversely affect subsequent operations, and the calibration accuracy tolerances specified in 4.4.1, 4.4.2, and 4.4.8 shall apply.

4.4.9.2 High temperature.--The tester shall be placed within a test chamber capable of maintaining an internal temperature of 71°C., plus or minus 2° C., for a period of 24 hours. At the end of the 24-hour period, the temperature shall be reduced to 55°C. and shall be maintained at this temperature for a minimum period of 4 hours. At the end of the 4-hour period and while at plus 55°C., plus or minus 2°C., the tester shall be subjected to the tests specified in 4.4.1, 4.4.2, and 4.4.8. The decade resistance values obtained shall not vary more than 0.4-percent from the initial-resistance values specified in tables III and IV. The emf values obtained shall not vary more than 0.08-mv between 0 and 20 mv, and not more than 0.12-mv between 20 and 50 mv, from the initial values specified in table VI. Resistance standards of 2, 8, and 22 ohms, maintained at room temperature, shall be measured by means of the thermocouple lead resistance measuring circuit. The error registered by adjustment No. 4 shall not differ by more than plus or minus 0.05-ohm from the true re-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

sistance values of the standards. After the tester has returned to room temperature, measurements shall be conducted to determine the calibration accuracy. There shall be no damage which would adversely affect subsequent operations, and the calibration accuracy tolerance specified in 4.4.1, 4.4.2, and 4.4.8 shall apply.

4.4.9.3 Fungus.--The tester shall be subjected to the fungus test, procedure I of MIL-STD-810. At the conclusion of the test, the equipment shall be carefully examined to ascertain that no fungus growth has occurred which will affect subsequent operation or cause deterioration or corrosion of any internal or external components. The calibration accuracy tolerances specified in 4.4.1, 4.4.2, and 4.4.8 shall apply.

4.4.9.4 Shock.--The tester shall not be damaged or subsequently fail to meet the tests specified in 4.4.1, 4.4.2, and 4.4.8 when subjected to impact shocks of 25 g. Each shock impulse shall have a time duration of 11 milliseconds, plus or minus 1 millisecond. The shocks shall be applied as follows:

(a) Horizontally, parrallel to the major horizontal axis (five shocks in each direction).

(b) Horizontally, at right angles to the major horizontal axis (five shocks in each direction).

(c) Vertically (five shocks in each direction).

4.4.10 Sealed meter.--The meter shall be immersed in tap water to the depth not exceeding 10 inches. The absolute pressure of the air above the water shall then be reduced to 2.5 inches Hg and maintained for 1 minute, or until air bubbles cease to be given off by the water, whichever is longer. There shall be no evidence of leakage as indicated by bubbles emanating from the interior of the meter. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as leaks.

4.4.11 Meter response.--Steady electric power, sufficient to produce momentary full scale deflection, shall be applied to the meter. The length of time, in seconds, required for the pointer to come to rest shall be noted. This test shall be repeated several times and the average values taken as the response time of the meter. The response time shall meet the requirements specified in 3.4.3.8.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

4.4.12 Resistive circuit capacity cycling (switches Nos. 2 and 8).--The calibration accuracy of the decade-resistance circuit shall be accurately determined in accordance with 4.4.1. Switches Nos. 2 and 8 shall then be operated mechanically for 10,000 cycles. One cycle of switch No. 2 shall consist of a complete rotation of the selector arm through 360 degrees. Switch No. 2 shall be rotated at a rate of 4 cycles per minute (cpm). One cycle of switch No. 8 shall consist of rotating the selector arm from the extreme-right position through the extreme-left position and back to the right position. Switch No. 8 shall be rotated at a rate of 10 cpm. Upon completion of the cycling test, the calibration accuracy shall again be accurately determined, and the changes on calibration shall not exceed the values obtained before cycling by more than 0.01-ohm at any test point.

4.4.13 Meter damping.--Electric power, sufficient to produce momentary full scale deflection, shall be applied to the meter. The maximum momentary deflection shall be observed and recorded. After the pointer has come to rest, the steady deflection shall be observed and recorded. When making this test, the circuit resistance shall be sufficiently high that no additional damping is added to the system. The damping factor shall meet the requirements specified in 3.4.3.3.

4.5 Preparation for Delivery.--

4.5.1 Preproduction package.--Levels A and B preproduction packages (see 4.2.1) shall be subjected to free fall drop tests in accordance with Method 5007 of FED-STD-101. Both procedure B and procedure E shall be performed. The drop height for level A shall be 30 inches; for level B, 22 inches.

4.5.2 Production.--Packaging, packing and marking of production items shall be inspected to assure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.--Testers shall be packaged level A, B or C, as specified (see 6.2).

5.1.1 Level A.--Each tester, immobilized as required in its carrying case, shall be cushioned in a snug-fitting wood-cleated plywood container conforming to overseas type, Grade B, of PPP-B-601. The preproduction package (see 4.2.1) shall protect the tester to 25Gs when tested as specified in 4.5.1 for level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

5.1.2 Level B.--Each tester, immobilized as required in its carrying case, shall be cushioned in a snug-fitting fiberboard container conforming to V3s, V4s or V3c of PPP-B-636. The preproduction package (see 4.2.1) shall protect the tester to 25Gs when tested as specified in 4.5.1 for level B.

5.1.3 Level C.--Each tester shall be packaged in a manner to prevent deterioration and damage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing.--Testers packaged per 5.1.1 or 5.1.2 shall not require packing. Testers packaged per 5.1.3 shall be packed in a manner to assure carrier acceptance and safe delivery to destination.

5.3. Marking.--In addition to any special requirements of the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.--The type N-3A tester is intended for use to determine the serviceability of electrical resistance and thermocouple-type temperature instruments used on aircraft in service.

6.2 Ordering data.--Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether first article approval is required (see 3.1.1).

(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

(d) Whether special marking is required (see 5.3).

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

6.3 Definitions.--

6.3.1 Sealed meter.--"Sealed meter" as defined herein is an instrument so constructed, protected, or treated that its successful operation and accuracy are not impaired when exposed to the environmental tests specified herein, and the meter may be operated successfully over a considerable period of time without damage to any of its parts.

6.3.2 Damping factor.--"Damping factor" as defined herein is the ratio of the steady deflection in angular degrees to the difference between the maximum momentary deflection and the steady deflection in angular degrees. For example: Let maximum momentary deflection be taken as 90 degrees. Let steady deflection be taken at 75 angular degrees. The 15 degrees is the difference between the two, and the damping factor is the ratio of 75.15, which equals 5.

6.3.3 Tapping.--"Gently tapped" as defined herein means the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

Custodian:

Army - AV

Preparing activity:

Army - AV

Review activities:

Army - AV

User activities:

Army -

Project No. 6695-A307

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

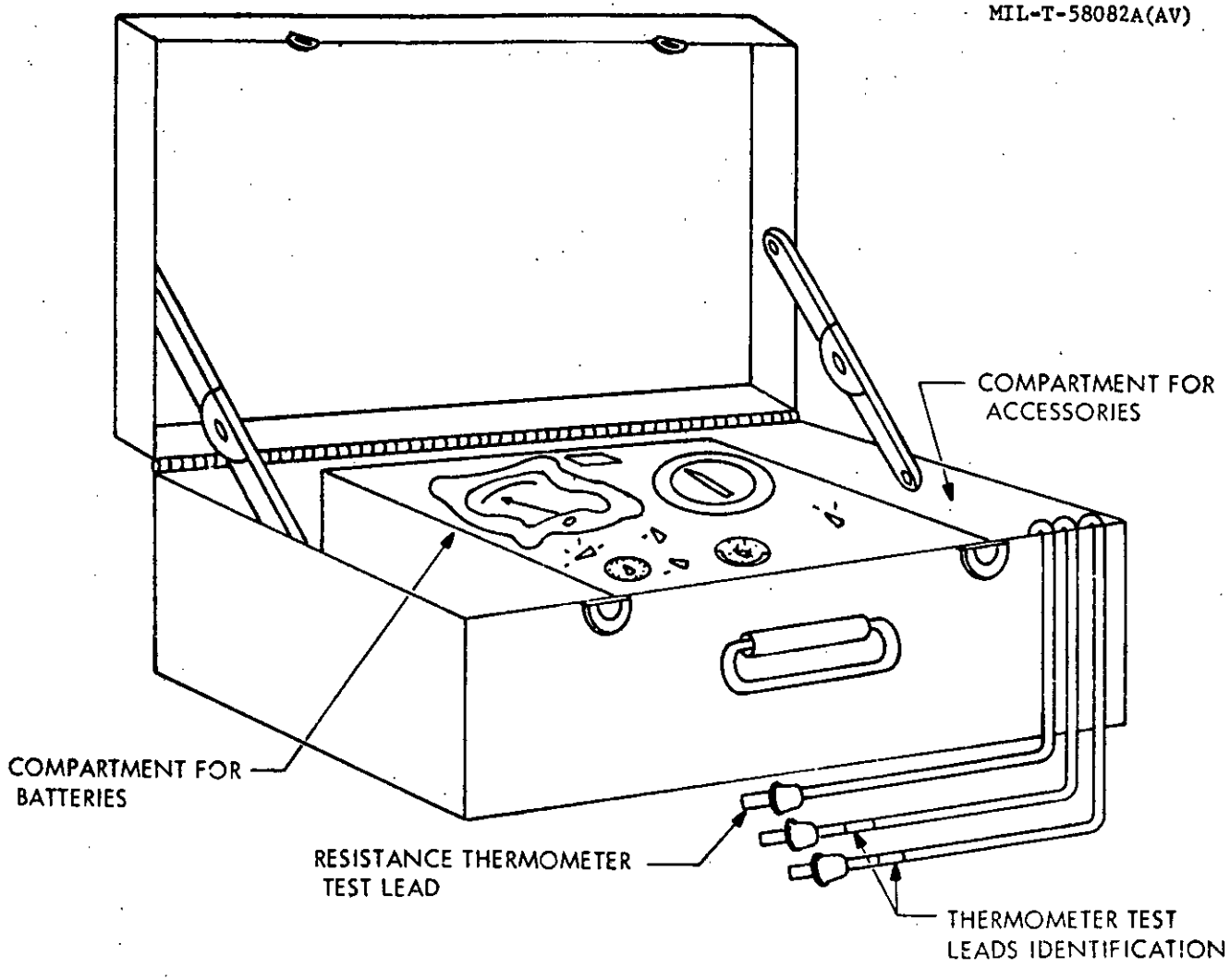


FIGURE 1. TESTER LAYOUT

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

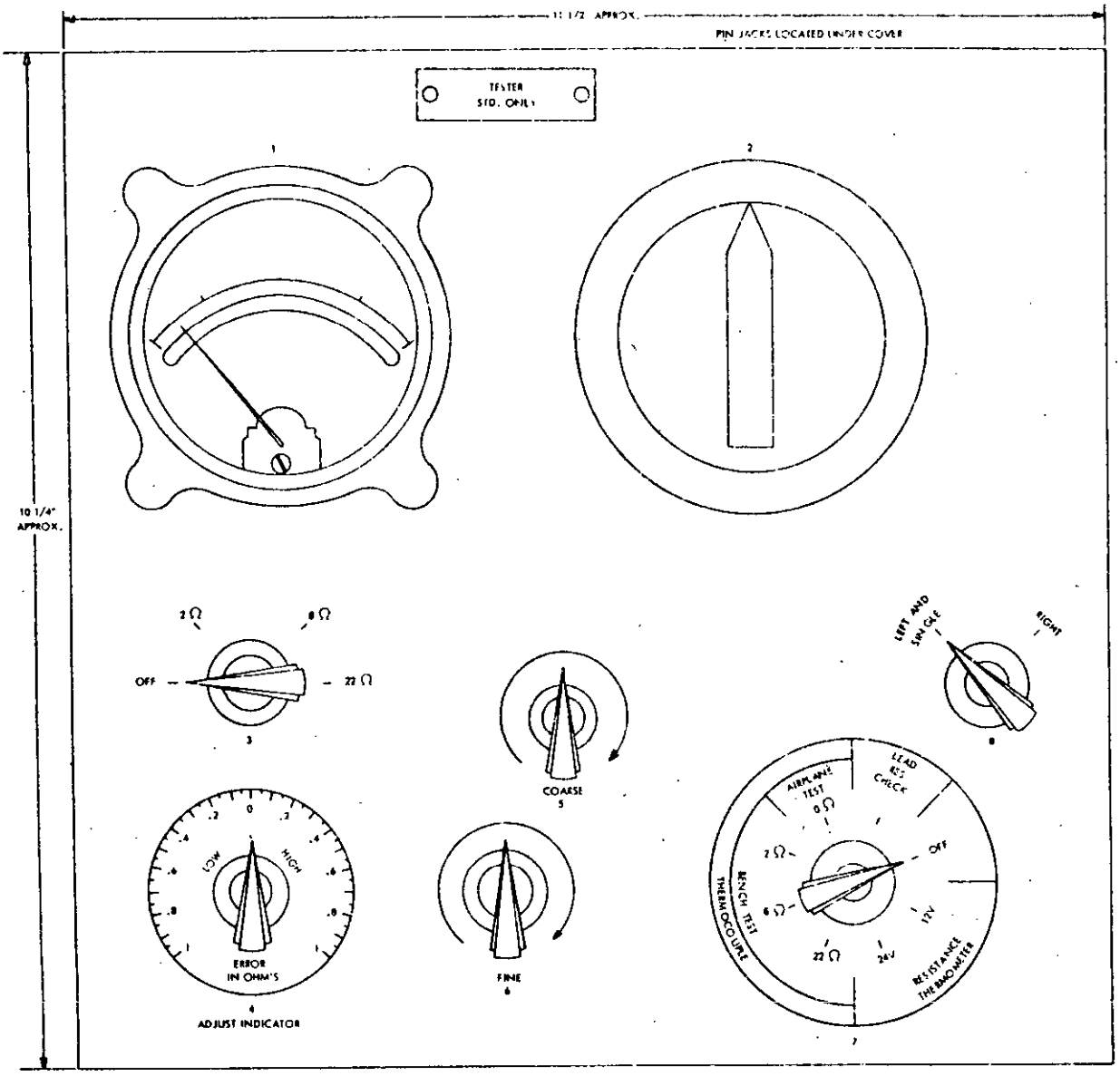


FIGURE 2. TEST PANEL

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

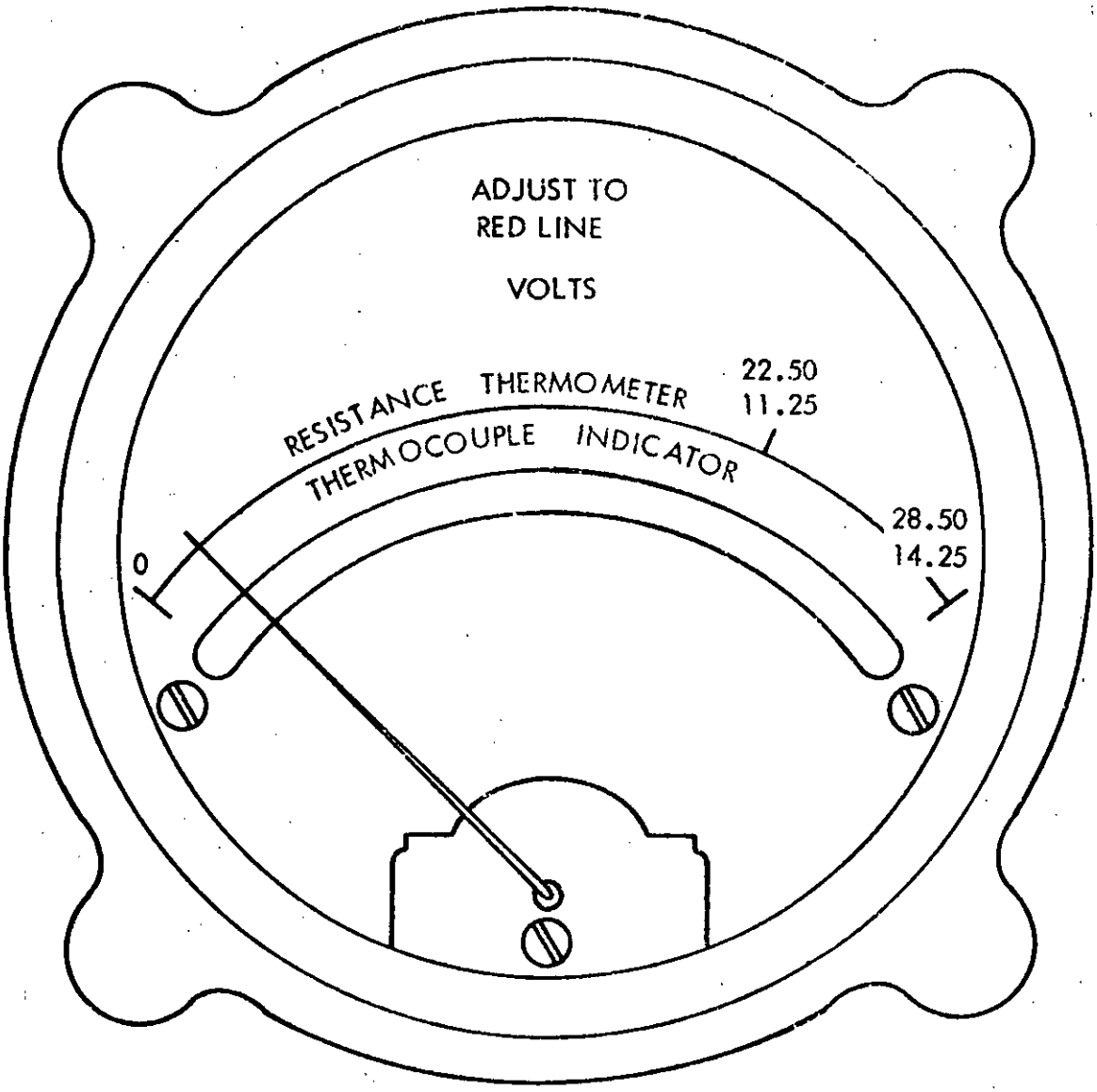
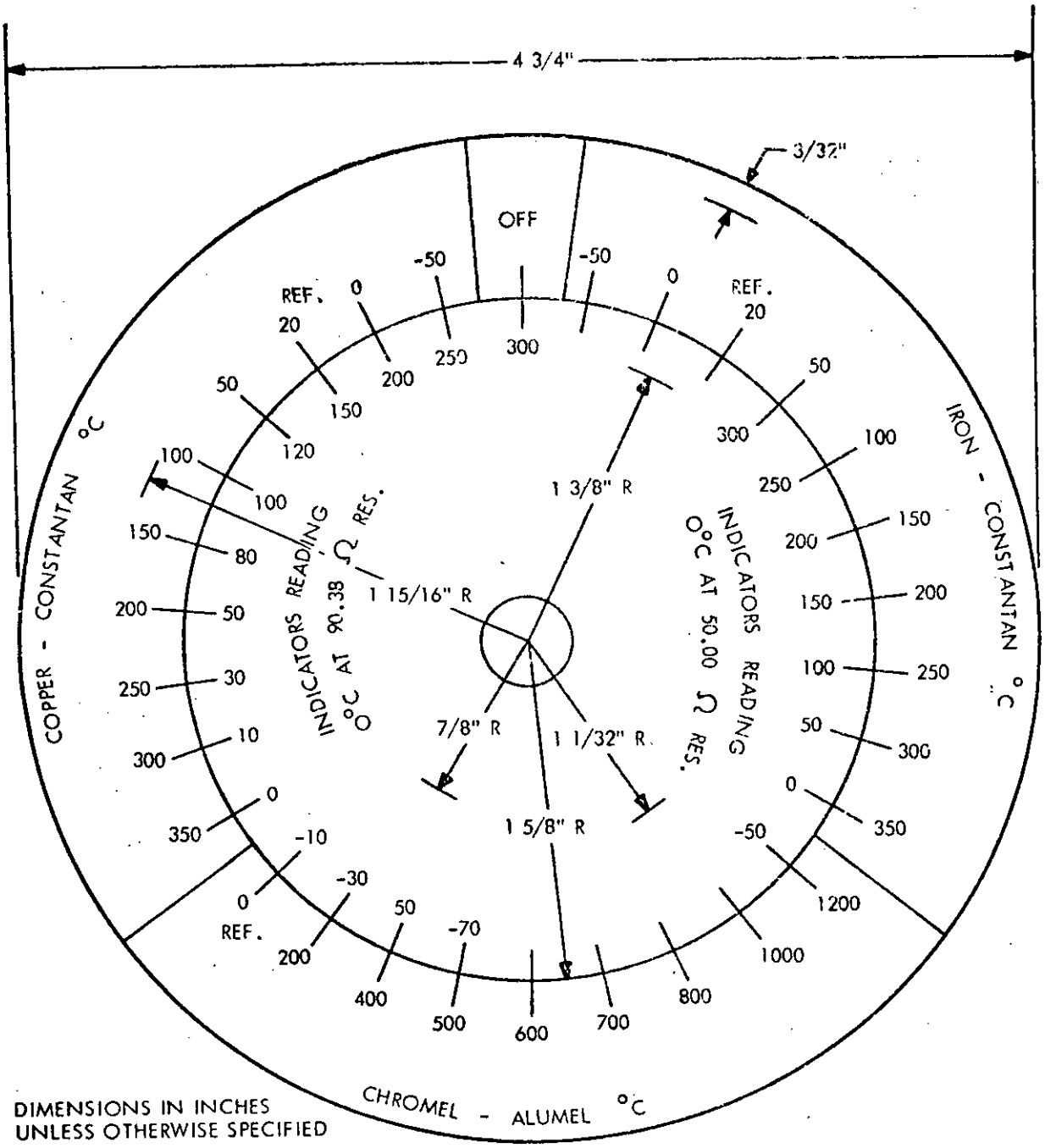


FIGURE 3. VOLTMETER

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED

TOLERANCES
FRACTIONS ± 1/64"

FIGURE 4. DIAL FOR TEMPERATURE SELECTOR SWITCH

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

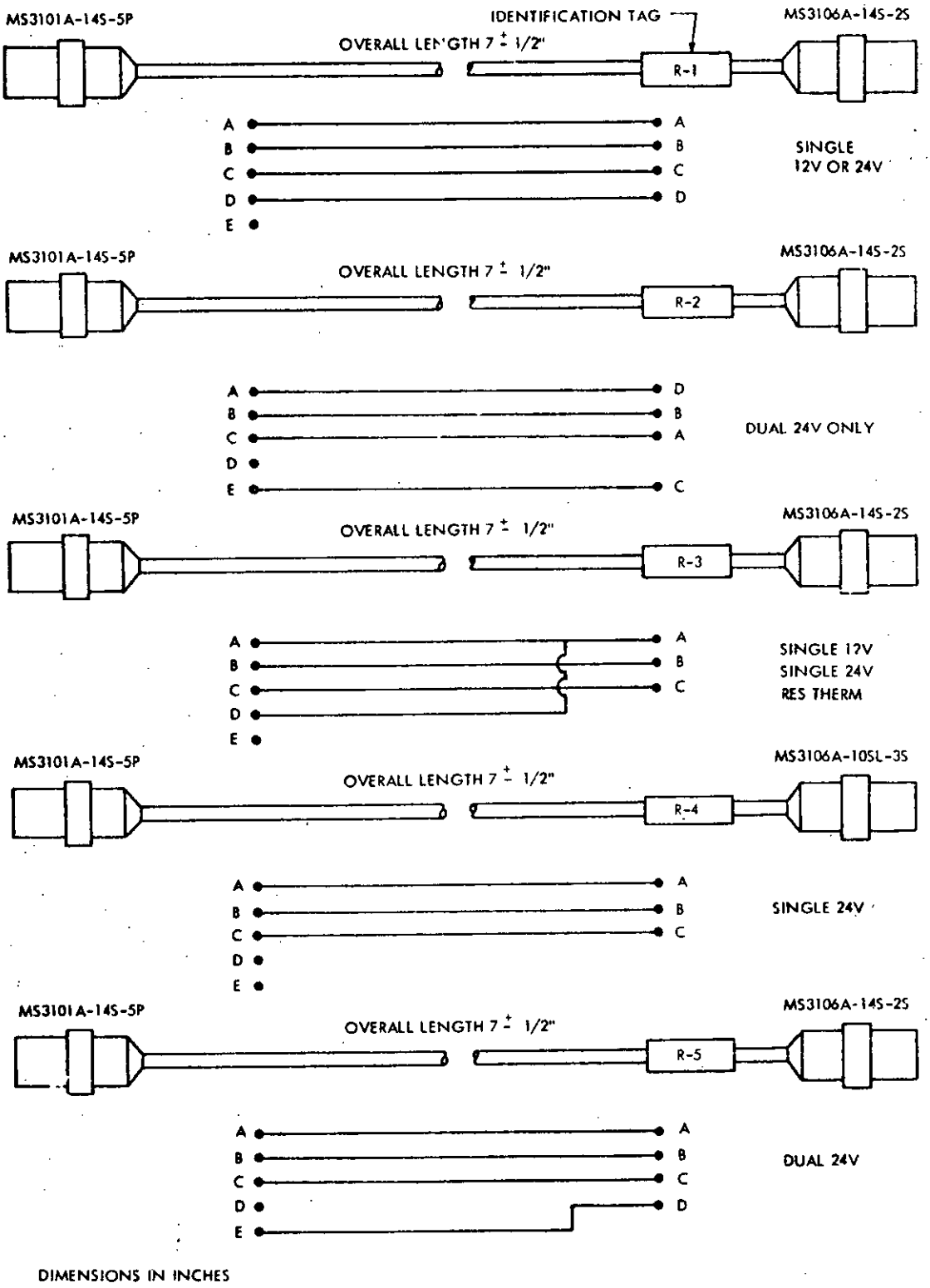


FIGURE 5. ADAPTER TEST LEADS

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

US Army Aviation Systems Command
AttN; AMSAV-ES
P.O. Box 209
St. Louis, MI 63166

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:00Z
Check the source to verify that this is the current version before use.